

Cancer Vaccines

Cancer

- Uncontrolled growth of cells
- Cancerous cells routinely arise in our body
- They are caused by mutations in some gene and hence express antigens that immune system can detect
- Immune system destroys them almost all the times
- Tumors are formed when immune system fails to destroy them

Cancer Prophylactic Vaccine

- Some cancer are caused by oncoviruses (e.g. cervical cancer)
- Vaccines against these viruses prevent such cancers
 - Hepatitis B vaccine to prevent liver cancer
 - Human papillomavirus vaccine that is responsible for ~70% of cervical cancers

Cancer Therapeutic Vaccine

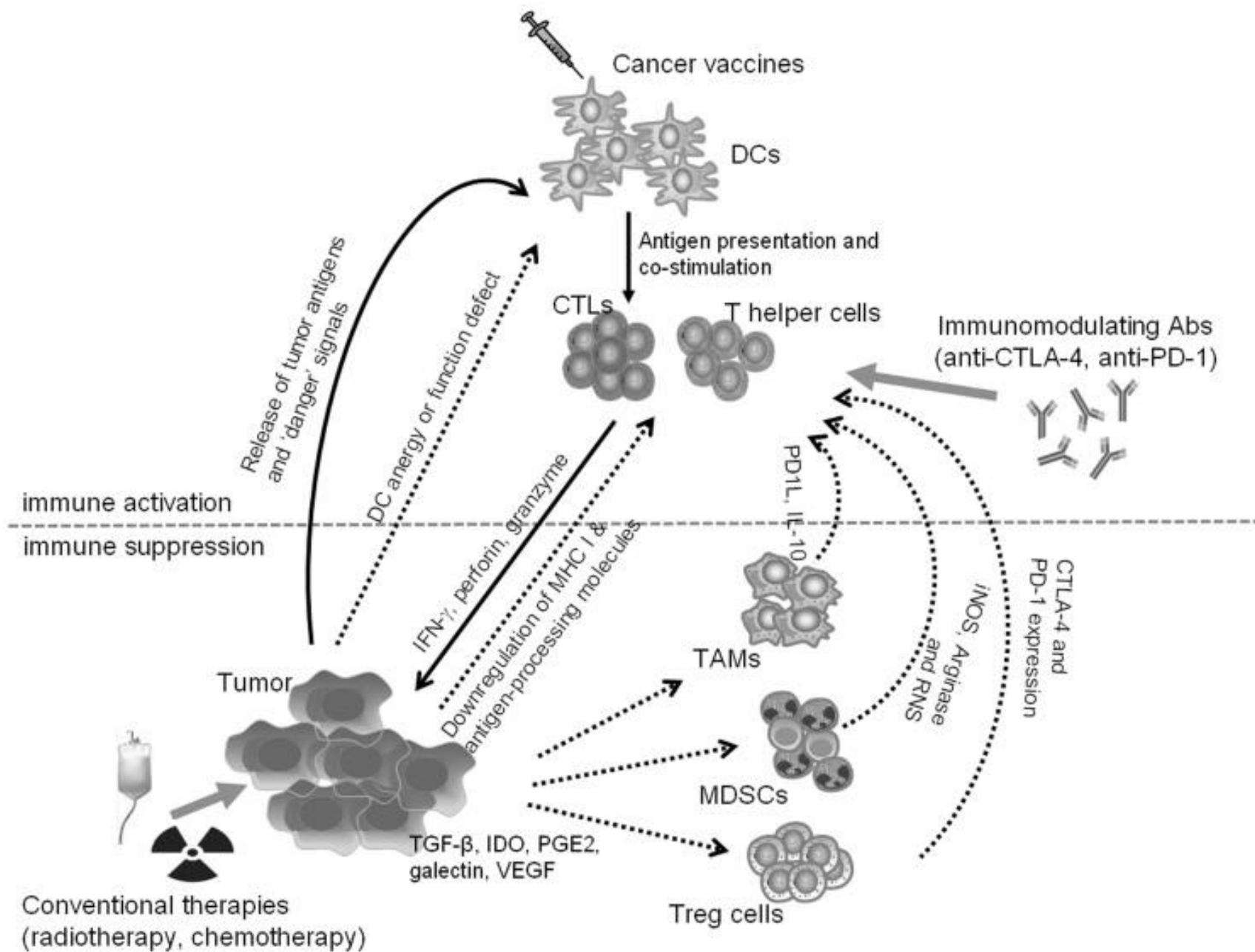
- Goal is to increase immune response against tumor antigens
- Hard to generate strong immune response due to similarity to the host tissue
- Tumors typically also have higher mutation rate and may develop several variants of tumor antigen

Therapeutic vaccines

- Autologous tumor cell vaccines
 - Isolate cells, lyse and inject with adjuvants
 - Presents entire spectrum of antigens
 - Transduction of stimulatory proteins such as IL-12, GM-CSF
 - Risk of autoimmunity
 - Need lots of tumor sample and may involve surgery

Potential tumour immune evasion mechanisms

- Physical barriers imposed by tumour architecture
- Induction of host myeloid and lymphoid suppressor cells
- Blocking T cell effector function
- Expression of proteases that interfere with cytotoxicity
- Expression of FAS-L
- Downregulation of cytokine receptors
- Resistance to apoptosis
- Evasion of T cell recognition
- Loss or mutation in target antigen
- Downregulation of MHC and other components important for antigen processing



Desirable properties of cancer vaccines

- **Targeting** antigen(s) preferentially or exclusively expressed on tumours but not on healthy tissue
- **Safety** — Delivery systems that are non-toxic to humans
- **Quantity** — Delivery systems capable of inducing sufficient numbers of effector T cells
- **Quality** — Delivery systems capable of inducing T cells of sufficient quality
 - Express T cell receptors with high avidity for target antigen
 - Have desired effector function (cytotoxicity, cytokine secretion)
 - Traffick to tumour sites
 - Duration of response sufficient to mediate tumour regression

Therapeutic vaccines

- Allogenic tumor cell vaccines
 - Uses few established human tumor cell lines
 - Limitless source and cost effective
 - Risk of autoimmunity
 - May not be effective at all

Sipuleucel-T (Provenge)

- Used clinically for prostate cancer since 2010
- Patients DCs are isolated
- Incubated with fusion protein (PA2024)
 - The [antigen prostatic acid phosphatase](#) (PAP), which is present in 95% of prostate cancer cells and
 - An immune signaling factor [granulocyte-macrophage colony stimulating factor](#) (GM-CSF) that helps the APCs to mature
- Injected back in the patients
- Improves life by 3 years

Stimuli-sensitive or Responsive Delivery Systems

- Magnetically controlled release systems
- Ultrasonically controlled systems
- Temperature sensitive hydrogels
- pH sensitive hydrogels
- Ligand (such as glucose) sensitive hydrogels
- Light sensitive hydrogels
- Other stimuli
 - Specific ion-sensitive hydrogels
 - Specific antigen-responsive hydrogels

Environmentally-Responsive/Smart Systems

pH-responsive systems for Intracellular Drug Delivery

Polymer design inspired by the mode of action of viruses

pH-sensitive polymers, such as poly(propylacrylic acid), PPAA, that become hydrophobic at the endosomal pHs and disrupt the endosomal membrane.

Endosomal pH: 5.5-6.5

In another design, membrane-disruptive polymers have acid-cleavable bonds that degrade within the endosome, exposing the backbone which is membrane-disruptive, leading to the release of the drug to the cytosol

Environmentally-Responsive/Smart Systems

pH-Responsive Enteric Coatings for oral delivery

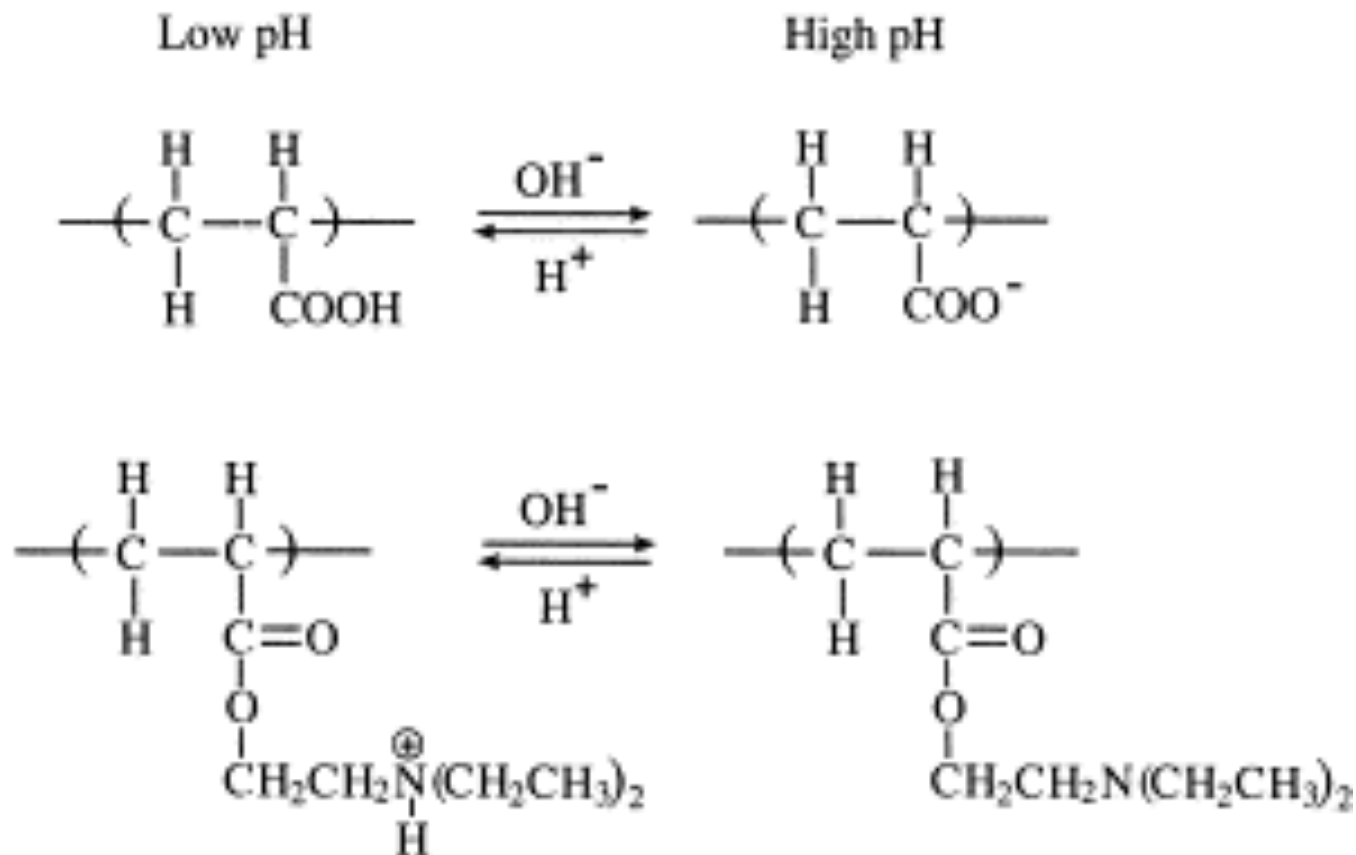
An enteric coating is a polymer that is insoluble at a low pH and soluble at physiologic pH.

The most common applications:

- (a) to protect acid-sensitive drugs from gastric juices,
- (b) to prevent the absorption of certain drugs in the stomach, and
- (c) to deliver drugs to the small intestine if that is their prime absorption site.

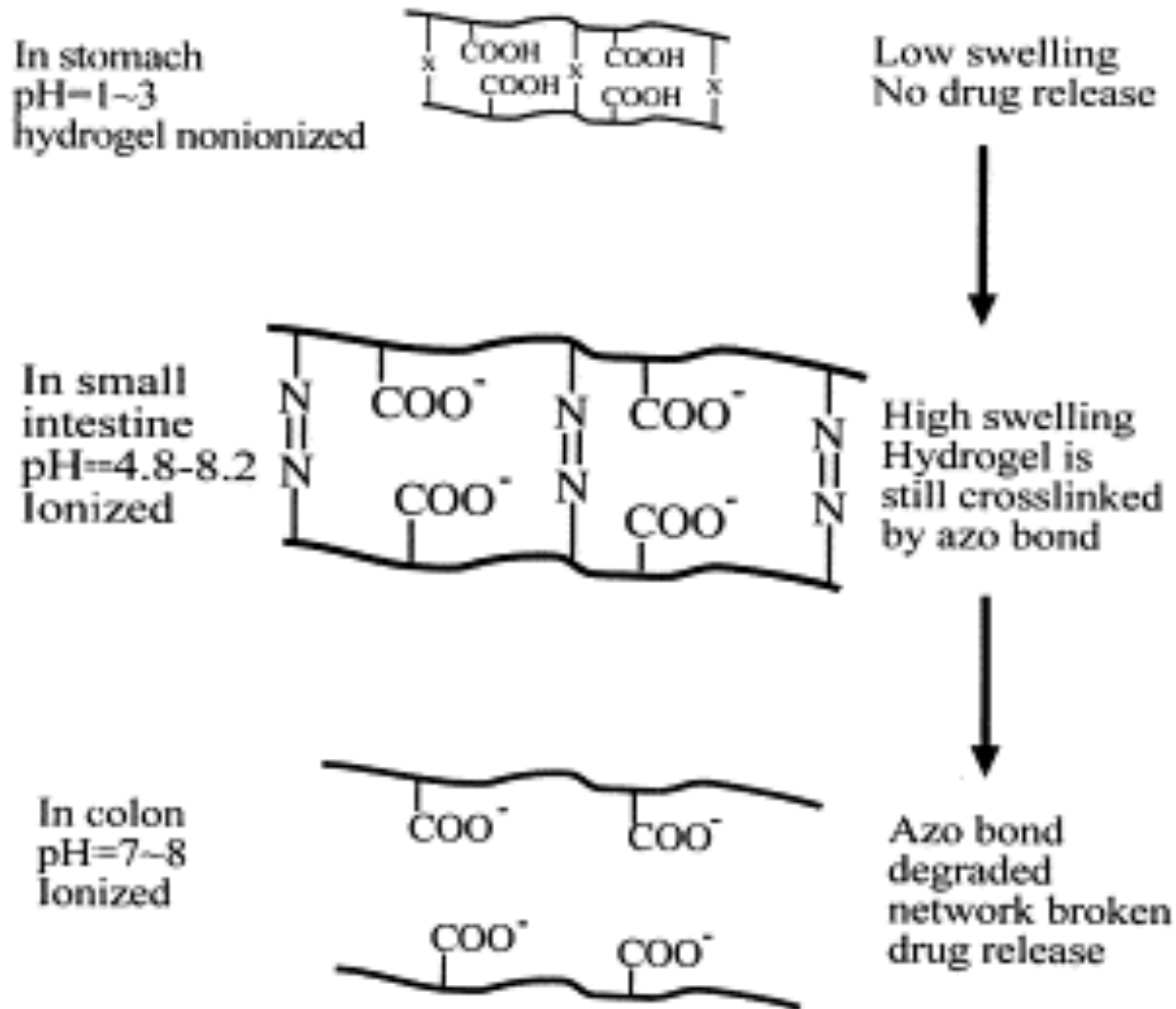
Modern enteric coatings are based on polycarboxylic acids that are hydrophobic and water-insoluble at a low pH become water soluble at a higher pH where the carboxyl groups ionize.

pH sensitive hydrogels



pH-dependent ionization of polyelectrolytes. Poly(acrylic acid) (top) and poly(*N,N'*-diethylaminoethyl methacrylate) (bottom).

pH sensitive, colon specific delivery



Azo reductase is produced by microbes present in colon

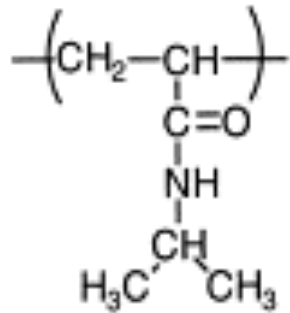
Temperature responsive hydrogels

- Most polymers increase their water-solubility as the temperature increases
- However, some polymers decrease their water-solubility as temperature increases (lower critical solution temperature (LCST))
- Hydrogels made of LCST polymers shrink as the temperature increases above the LCST
- Made of polymer chains that either possess moderately hydrophobic groups or contain a mixture of hydrophilic and hydrophobic segments
- At lower temperatures, hydrogen bonding between hydrophilic segments of the polymer chain and water molecules are dominant, leading to enhanced dissolution in water
- As the temperature increases, however, hydrophobic interactions among hydrophobic segments become strengthened, while hydrogen bonding becomes weaker. The net result is shrinking of the hydrogels due to inter-polymer chain association through hydrophobic interactions.

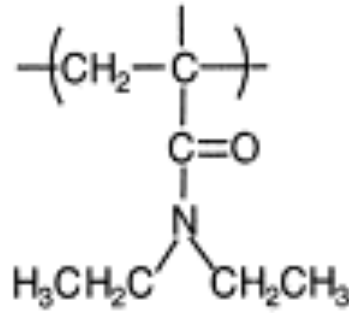
Temperature responsive hydrogels

- The LCST can be changed by adjusting the ratio of hydrophilic and hydrophobic segment of the polymer. More hydrophobic constituent lowers LCST
- Poly(*N*-isopropylacrylamide) (PNIPAAm) and Poly(*N,N*-diethylacrylamide) (PDEAAm) are the most extensively used because their LCST is in the range of 25-32°C
- Certain types of block copolymers made of poly(ethylene oxide) (PEO) and poly(propylene oxide) (PPO) also possess an inverse temperature sensitive property.

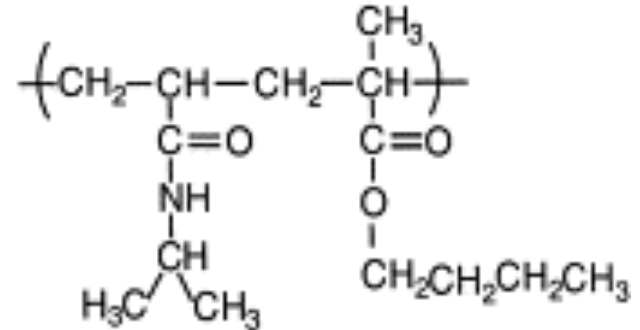
Temperature responsive hydrogels: Polymers



Poly(N-isopropylacrylamide)
(PNIAAm)

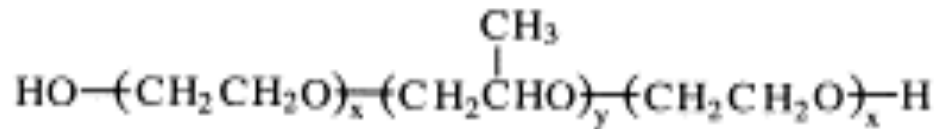


Poly(N,N-diethylacrylamide)
(PDEAAm)



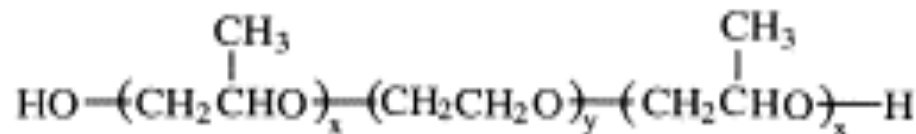
P(NIAAm-co-BMA)

Pluronic®



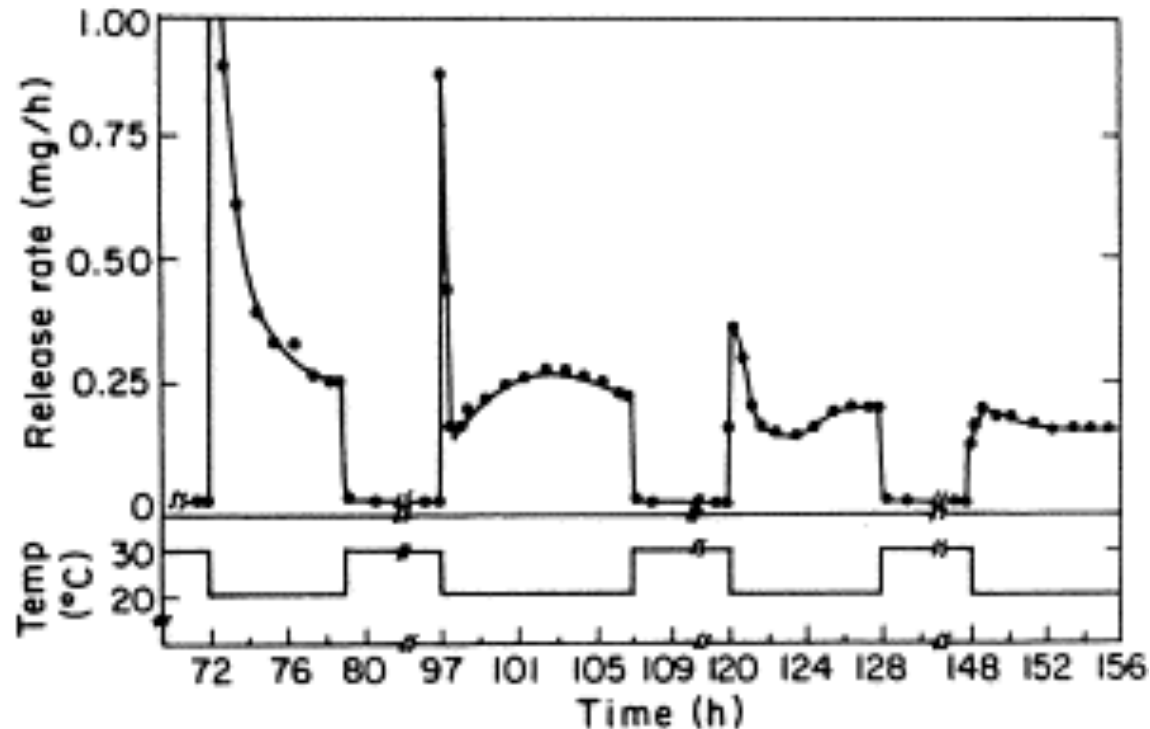
PEO-PPO-PEO

Pluronic® R



PPO-PEO-PPO

Temperature responsive hydrogels: Example



A pulsatile indomethacin release pattern was found to be regulated by temperature changes between 20°C and 30°C due to the reversible swelling properties of copolymers of *N*-isopropylacrylamide and butylmethacrylate

Temperature responsive Sol-Gel Hydrogels

- If the polymer chains in hydrogels are not covalently crosslinked, temperature-sensitive hydrogels may undergo solution-gelation phase transitions, instead of swelling-shrinking transitions
- Thermally reversible gels with inverse temperature dependence gels at higher temperatures
- A large number of PEO-PPO block copolymers are commercially available under the names of Pluronic[®] (or Poloxamers[®]) and Tetronics[®].

Temperature responsive hydrogels: Example



Advanced Drug Delivery Reviews 54 (2002) 37–51

Advanced
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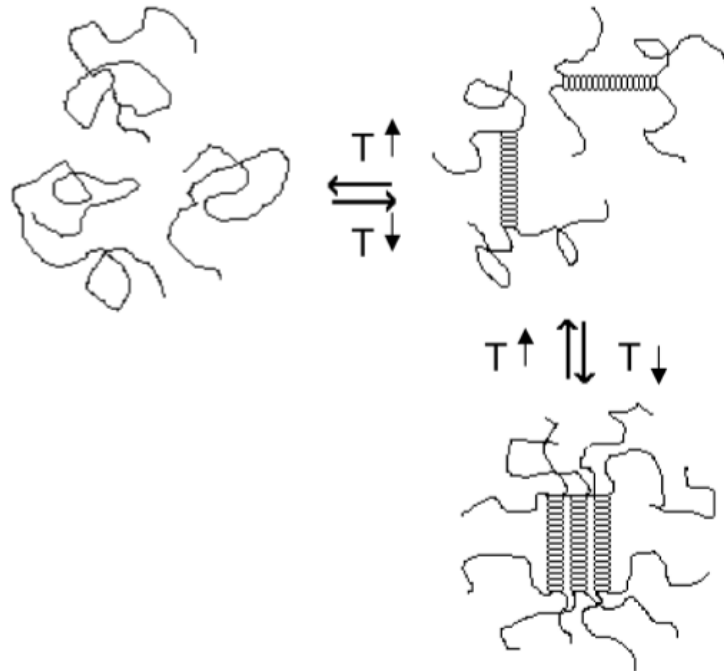
Thermosensitive sol–gel reversible hydrogels

Byeongmoon Jeong^a, Sung Wan Kim^b, You Han Bae^{b,*}

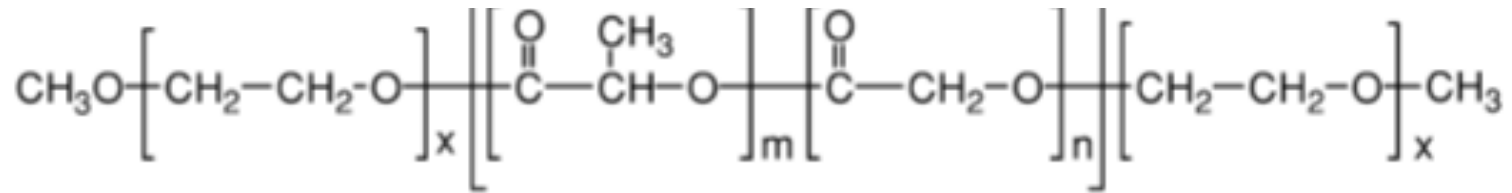
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^bCenter for Controlled Chemical Delivery, Department of Pharmaceutics and Pharmaceutical Chemistry, University of Utah, Salt Lake City, UT 84112, USA

Temperature responsive natural polymer gels



Thermo-reversible Hydrogels



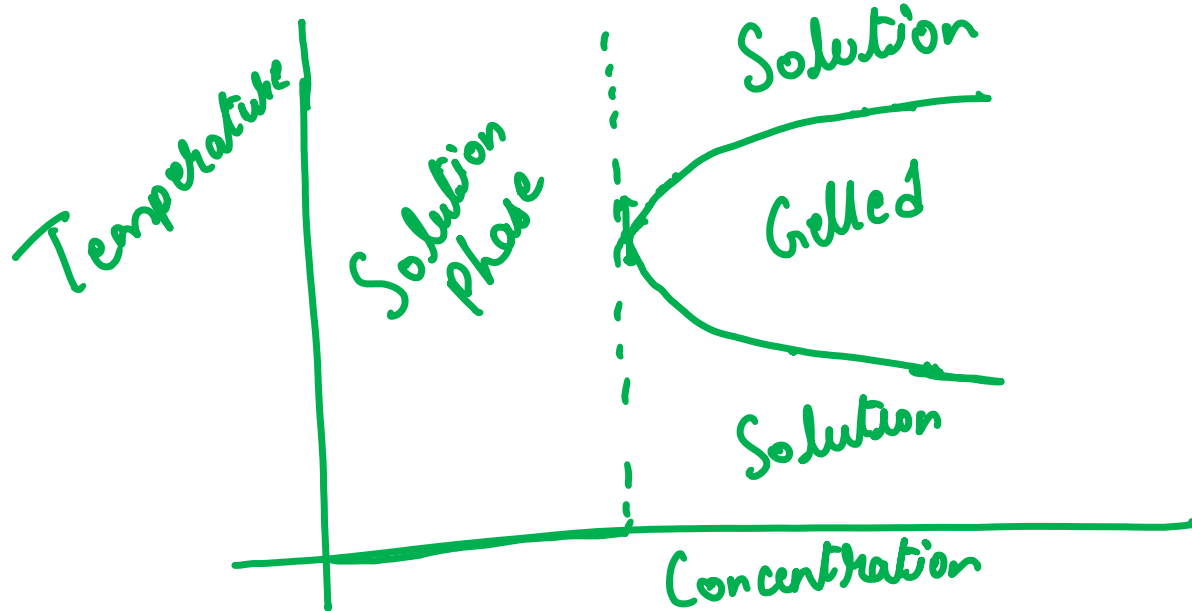
PEG block

PLGA block

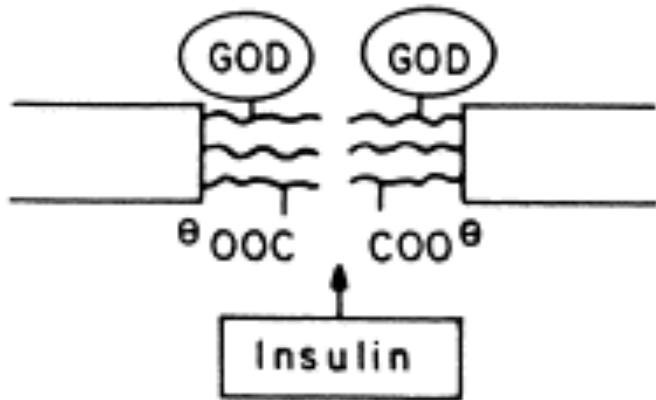
PEG block

Temperature vs concentration phase diagram?

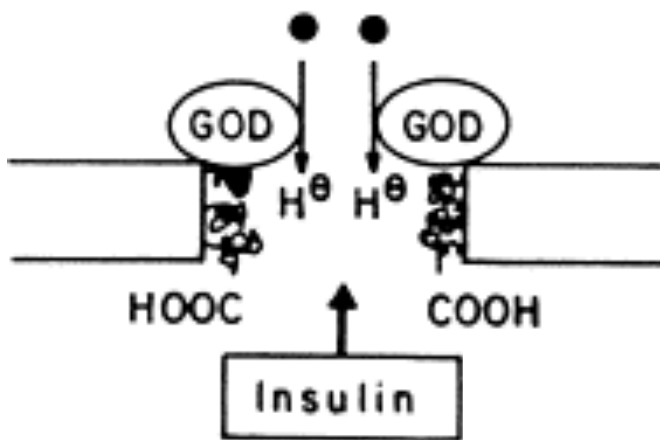
Sol - Gel
Solution - gelation



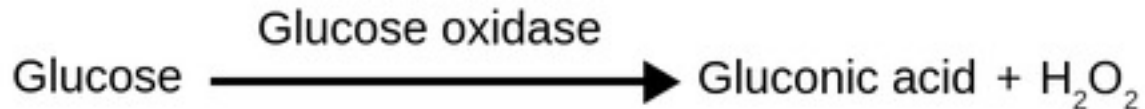
Glucose sensitive gels



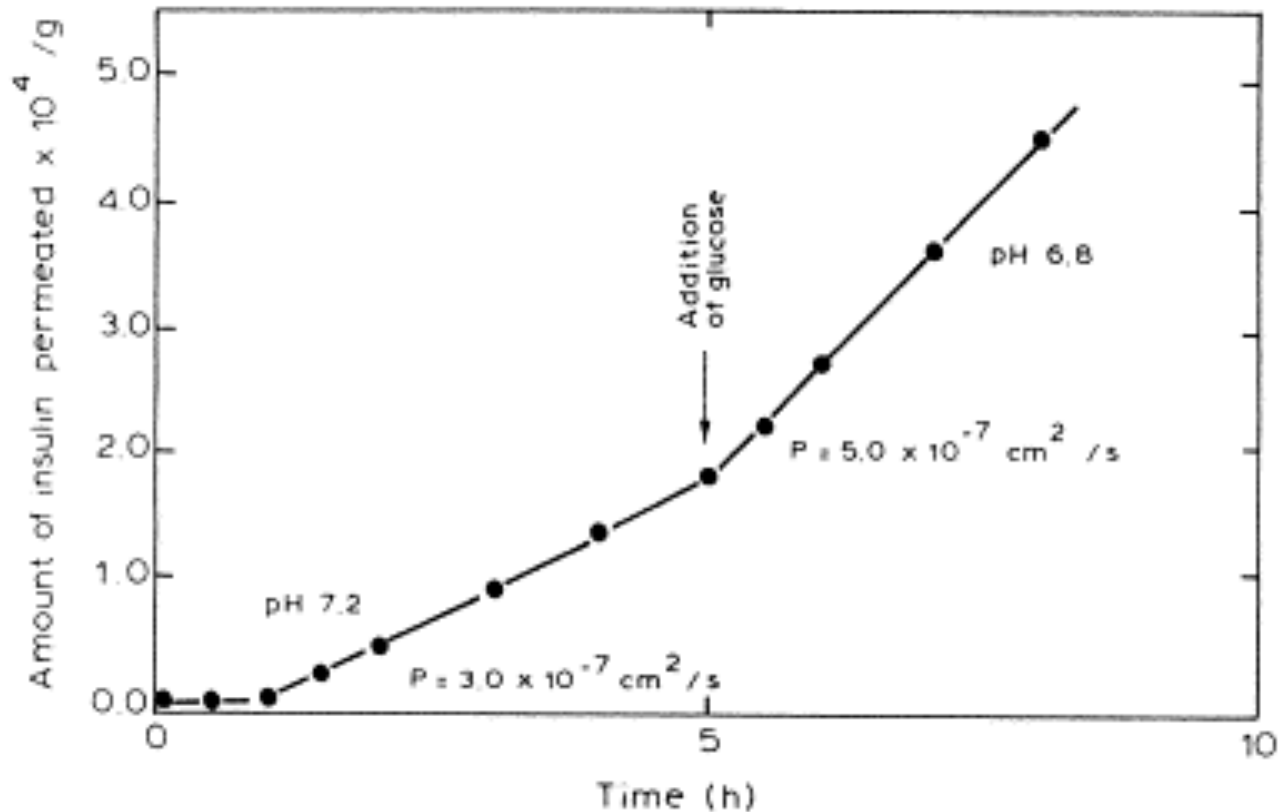
Principle of controlled-release system of insulin: glucose oxidase. Above: in the absence of glucose, the chains of poly(acrylic acid) grafts are rod-like, lowering the porosity of the membrane and suppressing insulin permeation. Below: in the presence of glucose, gluconic acid produced by glucose oxidase protonates the poly(acrylic acid), making the graft chains coil-like and opening the pores to enhance insulin permeation



●: glucose



Glucose sensitive gels

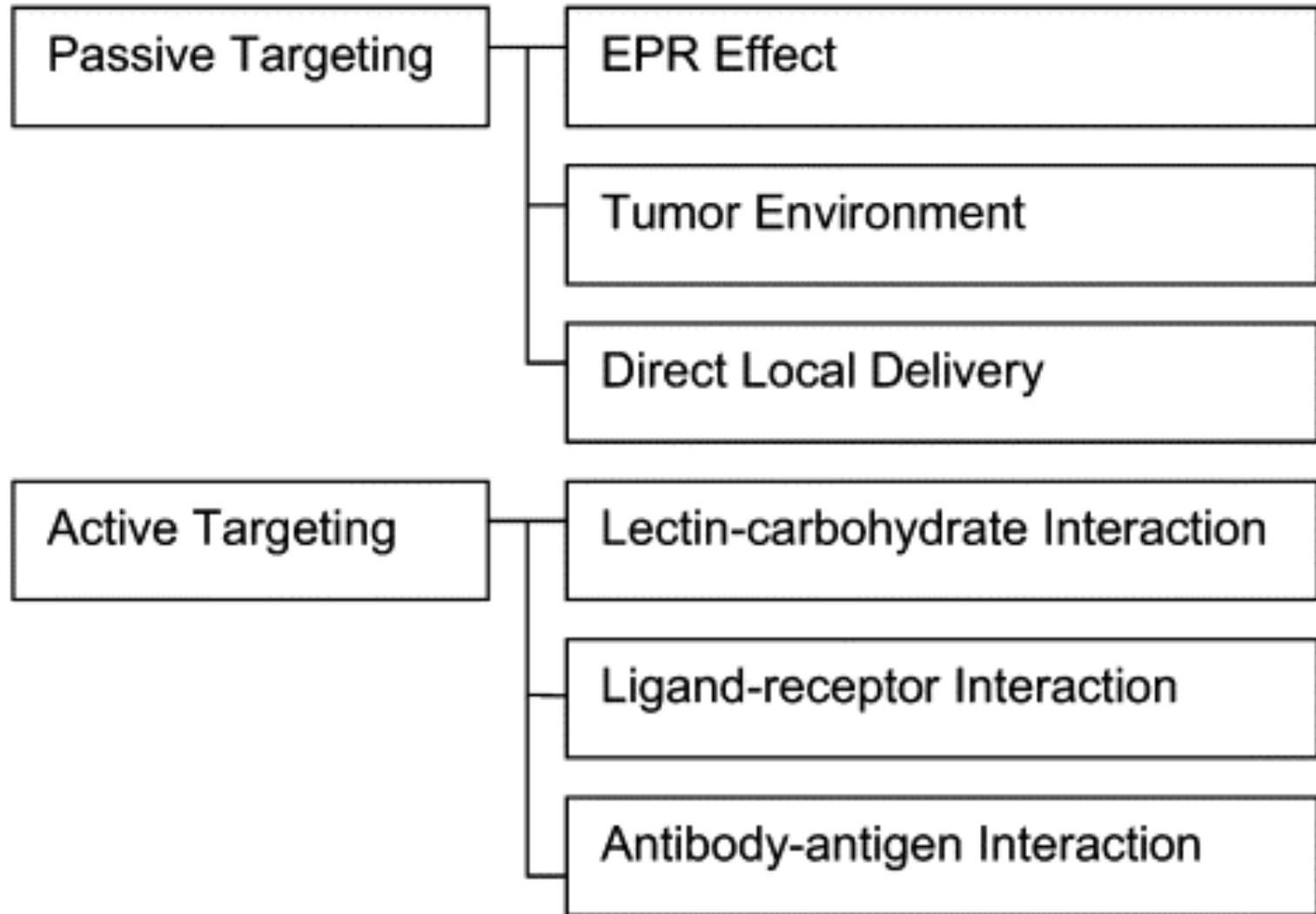


Permeation of insulin through poly(acrylic acid) grafted membrane in 0.1 M Tris-HCl-buffered solution. The concentration of added glucose is 0.2 M

Bio-inspired shielding strategies for particles and drugs

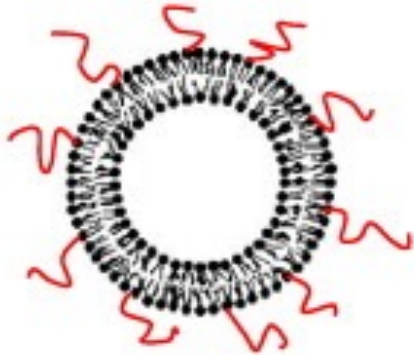
- Coating with Lipids
- Polymers
- Carbohydrates
- Proteins
- Give paper: **Bioinspired Shielding Strategies for Nanoparticle Drug Delivery Applications**

Targeted Drug Delivery System

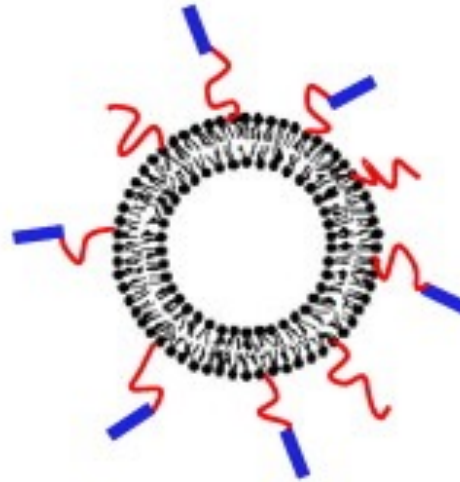


Targeted Drug Delivery System

Non-targeted nanoparticle



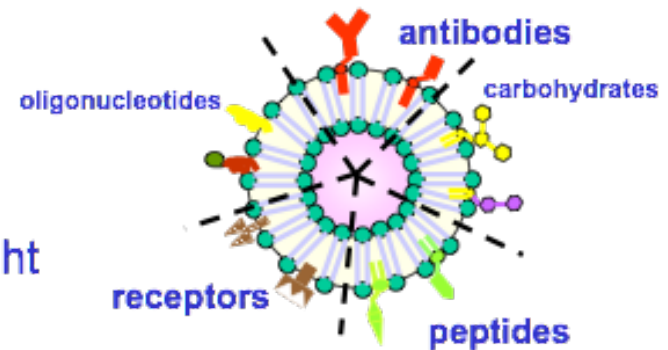
Targeted nanoparticle



— = Antibody (full)
— = Antibody (Fragment)
— = Small Molecule
— = Peptide
— = Aptamer

Targeting moieties:

- Antibodies
- Proteins
- Lipoproteins
- Hormones
- Charged molecules
- Polysaccharides
- Low-molecular-weight ligands



Tumor targeting

- Tumors upregulate their receptors to satisfy their need for fast growth and generation of new blood vessels
- Some molecules routinely upregulated in many tumor types are
 - Transferrin
 - EGF receptor
 - Folic acid
 - VEGF-A

Cons for targeted delivery systems

- Important to note that very few targeted systems are being used in clinics (e.g. trastuzumab emtansine (antibody-drug conjugate), for HER-2-positive breast cancer)
- Still debatable whether adding ligands for systemic administration has any therapeutic benefit
- Several issues such change in pharmacokinetics, masking by protein adsorption, ligand denaturation and accessibility of target

Clinically approved drug-delivery systems

Type of drug delivery system	Clinically approved drugs
Nanoparticles	Abraxane (Paclitaxel), Doxil (Doxorubicin), DaunoXome (Daunorubicin), Marqibo (Vincristine), MEPACT (Mifamurtide), Onivyde MM-398 (Irinotecan), ADYNOVATE (antihemophilic factor (recombinant) PEGylated), Estrasorb (estradiol), Ambisome (amphotericin B), Depocyte (cytarabine), Visudyne (Verteporfin)
Microparticle-based depots	Zmax (Azithromycin), Decapeptyl/Trelstar (Triptorelin), Vivitrol (Naltrexone), Arestin (Minocycline), Risperdal/Consta (Risperidone), Sandostatin LAR Depot (Octreotide), Nutropin Depot (Somatropin), Lupron Depot (Leuprolide), DepoCyt (Cytarabine), DepoDur (Morphine), Bydureon (Exenatide), Somatuline LA (Lanreotide), Zoladex (Goserelin), Suprefact Depot (Buselerin), Signifor (Pasireotide)
Transdermal materials and devices	Transderm-Scop (Scopolamine), Nitro-Dur (Nitroglycerin), Catapres-TTS (Clonidine), Estraderm (Estradiol), Duragesic (Fentanyl), Androderm (Testosterone), Combipatch (Estradiol with norethindrone), Lidoderm (Lidocaine), Climara Pro (Estradiol with levonorgestrel), Oxytrol (Oxybutynin), Synera (Lidocaine and tetracaine), Daytrana (Methylphenidate), Emsam (Selegiline), Neupro (Rotigotine), Exelon (Rivastigmine), Sancuso (Granisetron), Butrans (Buprenorphine), Ortho Evra (Estradiol and norelgestromin), Qutenza (Capsaicin), Flector (Diclofenac epolamine), NicoDerm/Habitrol/ProStep (Nicotine), Retin-A (Tretinoin), IONSYS (Fentanyl), SonoPrep (Lidocaine via ultrasound), Iontocaine (Lidocaine and epinephrine via iontophoresis), LidoSite (Lidocaine and epinephrine via iontophoresis)
Oral	Concerta (Methylphenidate), Ditropan XL (Oxybutynin), Teczem (Enalapril Diltiazem), Dilacor XR (Diltiazem), Covera-HS (Verapamil), DynaCirc CR (Isradipine), Minipress XL (Prazosin), Procardia XL (Nifedipine), Fortamet (Metformin), Altoprev (Lovastatin), Glucotrol XL (Glipizide), Invega (Paliperidone), Tegretol-XL (Carbamazepine), Allegra D (Pseudoephedrine and Fexofenadine), Efidac/24 (Pseudoephedrine and Brompheniramine or Chlorpheniramine), Volmax (Albuterol), Orenitram (Trepstinil), Sudafed 24 h (Pseudoephedrine), Exalgo (Hydromorphone), Vesanoid (Tretinoin), Syndros (Dronabinol), Venclexta (venetoclax), Farydak (panobinostat), Renagel (Sevelamer)
Pulmonary	Tudorza/Pressair (Aclidinium), Proventil HFA (Albuterol), Ventolin HFA (Albuterol), ProAir HFA (Albuterol), Combivent Respimat (Albuterol and ipratropium), DuoNeb (Albuterol and ipratropium), Brovana (Arformoterol), QVAR (Beclomethasone), Pulmicort Flexhaler (Budesonide), Symbicort (Budesonide and Formoterol), Alvesco (Ciclesonide), Breo/Ellipta (Fluticasone and vilanterol), Flovent HFA (Fluticasone), Flovent/Diskus (Fluticasone), Foradil/Aerolizer (Formoterol), Perforomist (Formoterol), Arcapta Neohaler (Indacaterol), Atrovent HFA (Ipratropium), Xopenex HFA (Levalbuterol), Asmanex/Twisthaler (Mometasone), Dulera (Mometasone and Formoterol), Serevent/Diskus (Salmeterol), ADVAIR Diskus (Salmeterol Fluticasone), ADVAIR HFA (Salmeterol Fluticasone), Spiriva/Handihaler (Tiotropium), Cayston (Aztreonam), Ventavis (Iloprost), Tyvaso (Trepstinil), TOBI Podhaler (Tobramycin), Afrezza (human insulin)
Implants	Vitrasert (Ganciclovir), Retisert (Fluocinolone), Ozurdex (Dexamethasone), Zoladex (Goserelin), Gliadel (Prolifeprosan and Carmustine), Vantas/Supprelin LA (Histrelin), Viadur (Leuprolide), Nexplanon (Etonogestrel), NuvaRing (Etonogestrel and ethinyl estradiol), Mirena/Norplant (Levonorgestrel), Paragard (Copper)

Steps towards approval

- **STEP ONE** in the marketing process is to make absolutely sure that the product that you wish to market is feasible
- **STEP TWO** is to determine how regulatory agencies (ICMR) may classify your device - which one of the classes the device may fall into. Classification identifies the level of regulatory control that is necessary to assure the safety and effectiveness of a medical device.
- **STEP THREE** is the development of data and/or information necessary to submit a marketing application, and to obtain clearance to market

GMP and QS Regulations

- Good Manufacturing Practice (GMP) and QS (Quality Systems) Regulations
- These regulations require that domestic or foreign manufacturers have a quality system for the design, manufacture, packaging, labeling, storage, installation, and servicing of finished medical devices intended for clinical trials and commercial distribution
- Anything used in humans should follow these guidelines

Investigation Phases in Clinical Trials

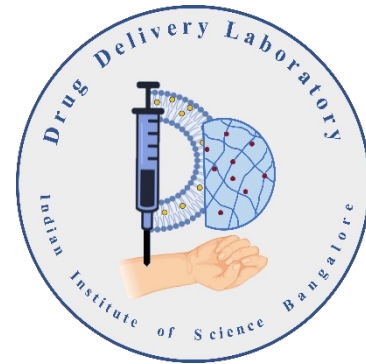
- Phase 1 Feasibility Study
 - Small numbers to assess product safety
 - Generally localized to one or two centers
- Phase 2 Study
 - Proper/safe dosing; potential efficacy
- Phase 3 Pivotal Study
 - Well controlled clinical trial supports safety/efficacy; leads to FDA application for premarket approval
 - Multi-center involvement
 - 90% of Phase III trials fail primarily due to lack of efficacy or due to unforeseen toxicity in specific human populations.

Overall process for approval

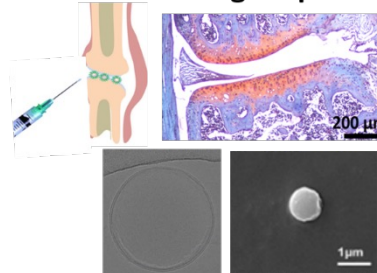
- New drugs need a full and comprehensive evaluation pathway (requires approval at each step)
 - Small animal trials (Animal Ethics committee)
 - Large animal trials (Large Animal/Primate Ethics committee)
 - Extensive toxicity screening
 - Human and follow-up clinical trials
- Repurposed drugs and drug delivery vehicles packed with pre-approved drugs and/or pre-evaluated excipients (e.g. PLGA and variants, phospholipids etc) have a much easier pathway

Interested Researchers: Drug Delivery Lab

- rachit@iisc.ac.in
- Bioengineering, TSH Building
- Positions:
 - Undergraduate semester projects
 - M. Tech projects
 - Ph.D.



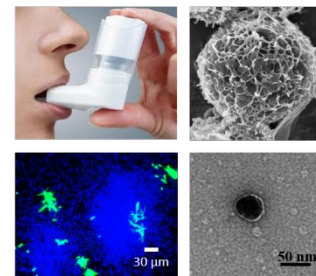
Osteoarthritis treatment
and cartilage repair



Pro-
Resolution
therapies

Alleviating
Cellular
Stress

TB models and drug
delivery



TB Drug
Delivery

TB in-
vitro
Models

Biomaterials
based
particle and
tissue
technologies